

Life of a Data Project

Assessment Briefs — Final Exam & Group Project

This document outlines the two assessments for the Life of a Data Project module. Both are designed to measure your ability to conduct a complete data project in line with the syllabus: from problem definition and data acquisition through cleaning, analysis, visualization, and storytelling. As first-year students, you are not expected to be experts — you are expected to show that you have understood and practiced every step of the pipeline. Please read both briefs carefully before you begin working.

1. Final Exam — Individual Assessment

At a glance

Format: individual, hands-on practical examination

Scope: the full data project lifecycle covered in the syllabus

Toolset: Python, Pandas, Matplotlib, Seaborn, BeautifulSoup, Selenium

Objective

The final exam is designed to assess your individual ability to conduct a complete data analysis, from data loading and preparation through exploration and visualization. You will be asked to show, on your own, that you have understood the methods, the tools, and the analytical mindset practiced throughout the course.

What you are expected to master

You should be comfortable operating across every stage of the data project lifecycle:

- **Problem framing:** translating a business question into a structured, data-oriented problem.
- **Data acquisition:** loading data from files, public sources, or APIs, and extracting data from the web when required (BeautifulSoup, Selenium).
- **Preprocessing and cleaning:** handling missing values, type conversions, duplicates, formats, and standardization of heterogeneous sources.
- **Feature engineering:** creating and transforming variables to enrich the dataset and support deeper analysis.
- **Exploratory analysis:** descriptive statistics, aggregations, correlations, and the identification of trends or anomalies.
- **Visualization:** building clear, relevant charts with Matplotlib and Seaborn to support your interpretation.

Expected level of autonomy

You will work individually on a dataset provided during the exam. Your code must be clean.

2. Group Project — Full Business Data Analysis

At a glance

Format: team project, delivered as a notebook and an oral presentation

Scope: end-to-end analysis of a business dataset, from raw data to recommendations

Grading: 40% notebook and analysis — 60% oral presentation

Objective

This group project puts you in the position of a data analyst working on a real business case. Your mission is to take ownership of a dataset, understand the business context around it, build a clear analytical plan, and deliver an analysis that leads to concrete recommendations. You will then present your work to a committee(your teacher), as you would in a professional setting.

What is expected of you

This is a first-year project, and you are not expected to deliver the work of a senior data analyst. What we do expect is that you take the case seriously and show clear, honest work at every step of the pipeline. The project will assess your ability to:

- Understand the business stakes behind the case and frame the right analytical questions.
- Understand your data deeply — its structure, its limitations, its story.
- Build a clear analytical plan before diving into code, and execute it rigorously.

- Conduct a complete and detailed analysis, from dataset loading and cleaning through exploration, to business-oriented findings and recommendations.
- Translate your results into a compelling narrative, supported by well-crafted visualizations.

A note on the use of AI

Storytelling matters, and so does technical craft. You must demonstrate that you are able to conduct a complete analysis on your own, without relying on AI assistants to carry the work for you. AI tools may support your workflow, but they cannot replace your own understanding of Pandas, visualization libraries, and analytical reasoning — all of which will be tested during the oral defense. Be ready to explain every line of code and every chart you produce.

Evaluation criteria

You will be assessed on four dimensions:

1. Your ability to understand the business case.
2. Your ability to understand and interrogate your data.
3. Your ability to conduct a complete analysis — from dataset loading and preparation, through exploration, to business recommendations.
4. Your ability to transmit your analysis orally, with clarity, structure, and impact.

Grading breakdown

- **40% — Notebook and written analysis:** code quality, methodological rigor, depth of exploration, quality of feature engineering, and reproducibility.
- **60% — Oral presentation:** clarity of the narrative, quality and relevance of visualizations, command of the content, and ability to answer the committee's questions.

Deliverables

- A clean, documented Python notebook containing the full analysis, with code, comments, and visualizations.
- An oral presentation delivered in front of the committee, framed as a business-style debrief: context, approach, findings, and recommendations.

Treat this project as a professional deliverable. The quality of your thinking should be visible in both your code and your words.